

• Register Map

Peripheral	Purpose	Base Address
RCC	Peripheral clock control	0x40023800
GPIO (A/B/C/D)	LEDs, buttons, SPI, and UART pins	0x40020000 – 0x40020C00
EXTI	External interrupt controller	0x40013C00
NVIC	Interrupt management	0xE000E100
TIM2	Floor movement simulation timer	0x40000000
TIM3	Door timing control	0x40000400
TIM4	PWM motor simulation	0x40000800
SPI1	Full-duplex SPI IPC communication	0x40013000
USART1	UART telemetry output	0x40011000

• PWM Frequency Calculation

The PWM signal was used to simulate the elevator motor speed using LED brightness control.

The PWM frequency equation is:

$$f_PWM = f_CLK / ((PSC + 1) \times (ARR + 1))$$

we use:

Clock Frequency = 16 MHz

PSC = 15

ARR = 99

Substituting in the equation:

$$f_PWM = 16,000,000 / ((15 + 1) \times (99 + 1))$$

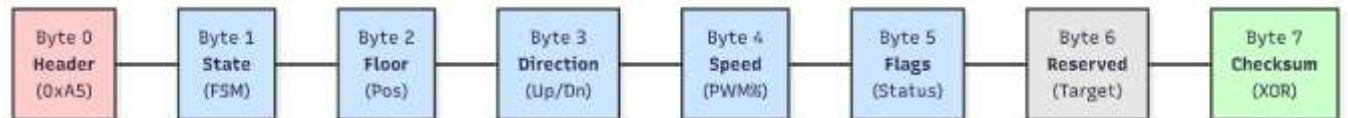
$$f_PWM = 16,000,000 / 1600$$

$$f_PWM = 10,000 \text{ Hz} = 10 \text{ kHz}$$

Therefore, the selected timer configuration generates a PWM signal with a frequency of approximately 10 kHz, which provides smooth LED dimming without visible flickering.

• SPI Packet Definition (8-Byte Frame)

The system uses a fixed-length 8-byte SPI packet for reliable full-duplex communication between the Master MCU and the Slave MCU.



- Packet Description:

- The Header byte (0xA5) is used for packet synchronization.
- The State field represents the current FSM state of the elevator.
- The Floor field indicates the current elevator position.
- The Direction field specifies the movement direction.
- The Speed field stores the PWM duty cycle percentage.
- The Flags byte contains emergency and door status indicators.
- The Reserved byte is used to transmit the target floor command from the Master MCU to the Slave MCU.
- The Checksum byte is generated using XOR across bytes 0 to 6 to detect corrupted SPI packets.

Checksum Equation:

$$\text{Checksum} = B0 \wedge B1 \wedge B2 \wedge B3 \wedge B4 \wedge B5 \wedge B6$$